

Martin Luther University of Halle-Wittenberg
Department of Economics
Chair of Econometrics

Econometrics II
3. Binary choice models

Christoph Wunder

Goals

- Introduce generalized linear models (GLMs) for binary outcomes
- Interpretation of GLMs: marginal effect and predictions
- Apply methods in R

Outline

③ Models for binary outcomes

3.1 Introduction

3.1.1 Random utility model

3.1.2 Latent variable model

3.2 Linear probability model (LPM)

3.3 Logit

3.4 Probit

3.5 Example: effectiveness of a new method of teaching

3.5.1 Estimation using maximum likelihood

3.5.2 Fitting a Bayesian model

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3.1.1 Random utility model

Cameron and Trivedi (2005), Ch. 14.4.2; Winkelmann and Boes (2006), Ch. 4.3

- Utility of two alternatives, U_0 and U_1 , depends on deterministic components, V_0 and V_1 , and random components, ϵ_0 and ϵ_1 :

$$U_0 = V_0 + \epsilon_0,$$

$$U_1 = V_1 + \epsilon_1,$$

where $V_0 = \mathbf{x}'_i \beta_0$ and $V_1 = \mathbf{x}'_i \beta_1$.

- Individual maximizes utility. We observe $y_i = 1$ if $U_1 > U_0$.
- The probability of observing $y_i = 1$ is

$$\begin{aligned} P(y_i = 1) &= P(U_1 > U_0) = P(V_1 + \epsilon_1 > V_0 + \epsilon_0) \\ &= P(\epsilon_0 - \epsilon_1 < V_1 - V_0) = F(V_1 - V_0). \end{aligned}$$

- $F(V_1 - V_0)$ is the cdf of $\epsilon_0 - \epsilon_1$.
- If $V_1 - V_0 = \mathbf{x}'_i \beta$, then $P(y_i = 1) = F(\mathbf{x}'_i \beta)$.

3.1.2 Latent variable model

Cameron and Trivedi (2005), Ch. 14.4.1; Winkelmann and Boes (2006), Ch. 4.3

- A latent variable, y^* , indicates an unobserved propensity.
- Model for latent variable:

$$y_i^* = \mathbf{x}_i' \boldsymbol{\beta} + u_i$$

- We observe:

$$y_i = \begin{cases} 1 & \text{if } y_i^* > 0, \\ 0 & \text{if } y_i^* \leq 0. \end{cases}$$

- The probability of observing $y_i = 1$ is

$$\begin{aligned} P(y_i = 1) &= P(y_i^* > 0) = P(\mathbf{x}_i' \boldsymbol{\beta} + u_i > 0) \\ &= P(-u_i < \mathbf{x}_i' \boldsymbol{\beta}) = F(\mathbf{x}_i' \boldsymbol{\beta}). \end{aligned}$$

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3.2 Linear probability model (LPM)

- LPM estimates response probabilities by a linear function $\mathbf{x}'_i\beta$:

$$P(y_i = 1) = F(\mathbf{x}'_i\beta) = \mathbf{x}'_i\beta$$

- The dependent variable y_i can take on only the values zero and one. The error term can take on only the values $1 - \mathbf{x}'_i\beta$ and $-\mathbf{x}'_i\beta$. Therefore, the error term is heteroskedastic:

$$\begin{aligned} \text{Var}(\epsilon_i|\mathbf{x}_i) &= (-\mathbf{x}'_i\beta)^2 P(y_i = 0|\mathbf{x}_i) + (1 - \mathbf{x}'_i\beta)^2 P(y_i = 1|\mathbf{x}_i) \\ &= (\mathbf{x}'_i\beta)(1 - \mathbf{x}'_i\beta) \end{aligned}$$

- Use robust s.e. or GLS estimation.

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3.3 Logit

- The logit link maps a probability, $p_j = P(y_j = 1)$, to a linear model:

$$\text{logit}(p_i) = \ln\left(\frac{p_i}{1-p_i}\right) = \mathbf{x}_i' \boldsymbol{\beta}$$

- The log-odds are defined as

$$\ln \left(\frac{p_i}{1 - p_i} \right) .$$

- The inverse logit transformation is the logistic function:

$$p_i = \frac{\exp(\mathbf{x}'_i \beta)}{1 + \exp(\mathbf{x}'_i \beta)}$$

Estimation by maximum likelihood

- Likelihood function:

$$\begin{aligned} L(\beta) &= \prod_{i=1}^N f_i(y_i | \beta, \mathbf{x}_i) \\ &= \prod_{i=1}^N p_i^{y_i} (1 - p_i)^{1-y_i} \\ &= \prod_{i=1}^N \left(\frac{\exp(\mathbf{x}_i' \beta)}{1 + \exp(\mathbf{x}_i' \beta)} \right)^{y_i} \left(\frac{1}{1 + \exp(\mathbf{x}_i' \beta)} \right)^{1-y_i} \end{aligned}$$

- Log-likelihood function

$$\ln L(\beta) = \sum_{i=1}^N \left\{ y_i \ln \left(\frac{p_i}{1 - p_i} \right) + \ln(1 - p_i) \right\}$$

- The R function `glm(..., family = binomial(link = "logit"))` fits a logit model by ML.

Interpretation of parameters

- Parameters are not equal to the marginal effects.
- Generally,

$$\frac{\partial p_i}{\partial x_k} = \frac{dF(\mathbf{x}'_i|\boldsymbol{\beta})}{d(\mathbf{x}'_i|\boldsymbol{\beta})} \frac{\partial \mathbf{x}'_i|\boldsymbol{\beta}}{\partial x_k} = f(\mathbf{x}'_i|\boldsymbol{\beta})\beta_k,$$

where f is the first derivative.

- For the logit model,

$$\begin{aligned} \frac{\partial p_i}{\partial x_k} &= \left(\frac{\exp(\mathbf{x}'_i \boldsymbol{\beta})}{(1 + \exp(\mathbf{x}'_i \boldsymbol{\beta}))^2} \right) \beta_k \\ &= \left(\frac{\exp(\mathbf{x}'_i \boldsymbol{\beta})}{1 + \exp(\mathbf{x}'_i \boldsymbol{\beta})} \right) \left(\frac{1}{1 + \exp(\mathbf{x}'_i \boldsymbol{\beta})} \right) \beta_k = F(\mathbf{x}'_i \boldsymbol{\beta})[1 - F(\mathbf{x}'_i \boldsymbol{\beta})]\beta_k. \end{aligned}$$

- The sign of the marginal effect is equal to the sign of β_k .
- The effect is largest for $\mathbf{x}'_i\beta = 0$.
- The effect varies among individuals.
- Average marginal effect:

$$N^{-1} \sum_{i=1}^N f(\mathbf{x}'_i \boldsymbol{\beta}) \beta_k$$

- Marginal effect at the average:

$$f(\bar{\mathbf{x}}'\boldsymbol{\beta})\beta_k$$

- For discrete or binary explanatory variables, we compute

$$\Delta p_j = F(\mathbf{x}'_j \boldsymbol{\beta} + \Delta x_{jk} \beta_k) - F(\mathbf{x}'_j \boldsymbol{\beta}).$$

Logit

- A measure of relative effect for a logit model is the proportional change in odds (odds ratio).
- The odds are

$$\frac{p_i}{1 - p_i} = \exp(\mathbf{x}_i' \boldsymbol{\beta}).$$

- The odds ratio is

$$\frac{\exp(\mathbf{x}_i' \boldsymbol{\beta} + \Delta x_{ik} \beta_k)}{\exp(\mathbf{x}_i' \boldsymbol{\beta})} = \exp(\Delta x_{ik} \beta_k).$$

- For a one unit change in x_{ik} , the odds ratio is $\exp(\beta_k)$ and the percentage change in odds is

$$[\exp(\beta_k) - 1]100.$$

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3.4 Probit

- The probit link is a common alternative:

$$\Phi^{-1}(p_i) = \mathbf{x}'_i \boldsymbol{\beta}.$$

- The probit model assumes that

$$p_i = F(\mathbf{x}'_i \boldsymbol{\beta}) = \Phi(\mathbf{x}'_i \boldsymbol{\beta}).$$

- The probit model can be motivated by a linear model for a latent continuous variable y_i^* (e.g. Winkelmann and Boes 2006, Ch. 4.2.3).

Estimation by maximum likelihood

- Likelihood function:

$$\begin{aligned}
 L(\beta) &= \prod_{i=1}^N f_i(y_i | \beta, \mathbf{x}_i) \\
 &= \prod_{i=1}^N p_i^{y_i} (1 - p_i)^{1-y_i} \\
 &= \prod_{i=1}^N \left[\Phi \left(\frac{\mathbf{x}_i' \beta}{\sigma} \right) \right]^{y_i} \left[1 - \Phi \left(\frac{\mathbf{x}_i' \beta}{\sigma} \right) \right]^{1-y_i}
 \end{aligned}$$

- Note that only the ratio β/σ is identified. Typically, we assume $\sigma = 1$.
- The R function `glm(..., family = binomial(link = "probit"))` fits a probit model by ML.

Probit

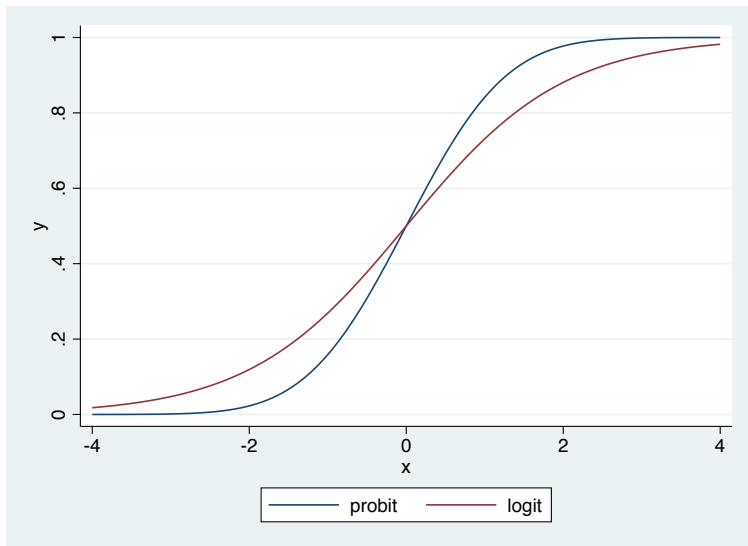


Figure: Comparison of logit and probit

Heteroskedasticity

- In the linear model, the OLS estimator is unbiased and consistent in the presence of heteroskedasticity. Robust s.e. are an easy-to-apply correction.
- In nonlinear models, the ML estimator is generally inconsistent when heteroskedasticity is related to explanatory variables (Greene 2011, Ch. 17.3.1).
- Case study on the [Stata blog](#).

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Example: effectiveness of a new method of teaching

3.5 Example: effectiveness of a new method of teaching

Greene (2011), Ch. 17.3.2

- Research question: does a new method of teaching economics, the Personalized System of Instruction (PSI), significantly influenced performance in later courses?
- Outcome:

$$\text{grade} = \begin{cases} 1 & \text{if student's grade improved,} \\ 0 & \text{otherwise.} \end{cases}$$

- Explanatory variables:
 - tuce: score on a test that indicates entering knowledge of the material
 - gpa: grade point average
 - psi: participation in program (0/1)

Example: effectiveness of a new method of teaching

3.5.1 Estimation using maximum likelihood

```
d <- read.csv("data/Greene_TableF14-1.csv",
              header = TRUE, sep = ",")
glimpse(d, width = 60)

## Rows: 32
## Columns: 5
## $ OBS    <int> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 1~
## $ GPA    <dbl> 2.66, 2.89, 3.28, 2.92, 4.00, 2.86, 2.76, 2.~
## $ TUCE   <int> 20, 22, 24, 12, 21, 17, 17, 21, 25, 29, 20, ~
## $ PSI    <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
## $ GRADE  <dbl> 0, 0, 0, 0, 1, 0, 0, 0, 0, 1, 0, 0, 0, 1, 0, ~
```


Example: effectiveness of a new method of teaching

Descriptive statistics

Variable	Mean	SD	Min	Max
GRADE	0.34	0.48	0.00	1.00
TUCE	21.94	3.90	12.00	29
GPA	3.12	0.47	2.06	4.00
PSI	0.44	0.50	0.00	1

Example: effectiveness of a new method of teaching

```
m1 <- glm( GRADE ~ GPA + TUCE + PSI,
            family = binomial(link = "logit"),
            data = d)

m2 <- glm( GRADE ~ GPA + TUCE + PSI,
            family = binomial(link = "probit"),
            data = d)

m3 <- glm( GRADE ~ GPA + TUCE + PSI,
            family = gaussian(link = "identity"),
            data = d)
```

Example: effectiveness of a new method of teaching

summary(m1)

```
##
## Call:
## glm(formula = GRADE ~ GPA + TUCE + PSI, family = binomial(link = "logit"),
##      data = d)
##
## Deviance Residuals:
##      Min       1Q   Median       3Q      Max
## -1.9551  -0.6453  -0.2570   0.5888   2.0966
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) -13.02135    4.93127  -2.641  0.00828 **
## GPA          2.82611    1.26293   2.238  0.02524 *
## TUCE         0.09516    0.14155   0.672  0.50143
## PSI          2.37869    1.06456   2.234  0.02545 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 41.183  on 31  degrees of freedom
## Residual deviance: 25.779  on 28  degrees of freedom
## AIC: 33.779
##
## Number of Fisher Scoring iterations: 5
```

Example: effectiveness of a new method of teaching

Table: Estimates from three models

	Logit		Probit		LPM	
	Beta	S.E.	Beta	S.E.	Beta	S.E.
(Intercept)	-13.02	4.93	-7.45	2.57	-1.50	0.47
GPA	2.83	1.26	1.63	0.69	0.46	0.14
TUCE	0.10	0.14	0.05	0.08	0.01	0.02
PSI	2.38	1.06	1.43	0.59	0.38	0.14

Note: LPM with robust s.e.

- Estimates are substantially different.
- Approximately, $\hat{\beta}_{LPM} \approx 0.25\hat{\beta}_{logit} \approx 0.4\hat{\beta}_{probit}$ (for slopes).

Example: effectiveness of a new method of teaching

Average marginal effects

Compute average marginal effect for GPA:

```
d$me.ml <- dlogis( cbind(1, d$GPA, d$TUCE, d$PSI ) %*%
  coef(ml) ) * coef(ml) [ "GPA" ]
mean(d$me.ml)

## [1] 0.3625808
```

```
library(margins)
ame.ml <- margins( ml, data = d )
ame.ml

## Average marginal effects
## glm(formula = GRADE ~ GPA + TUCE + PSI, family = binomial(link =
"logit"),      data = d)

##      GPA      TUCE      PSI
## 0.3626 0.01221 0.3052
```

Example: effectiveness of a new method of teaching

Marginal effects at means

Compute marginal effect for GPA at average values (means) of covariates:

```
my.mea.ml <- dlogis(
  cbind(1, mean(d$GPA), mean(d$TUCE), mean(d$PSI) ) %*%
                                             coef(ml) ) * coef(ml) [ "GPA" ]
round( my.mea.ml, 2 )

##          [,1]
## [1,] 0.53
```

```
mea.ml <- margins( ml, data = d, at = list( GPA = mean(d$GPA),
      TUCE = mean(d$TUCE), PSI = mean(d$PSI) ) )
mea.ml

## Average marginal effects at specified values
## glm(formula = GRADE ~ GPA + TUCE + PSI, family = binomial(link =
"logit"),      data = d)

##   at (GPA) at (TUCE) at (PSI)   GPA   TUCE   PSI
##    3.117    21.94   0.4375 0.5339 0.01798 0.4493
```

Example: effectiveness of a new method of teaching

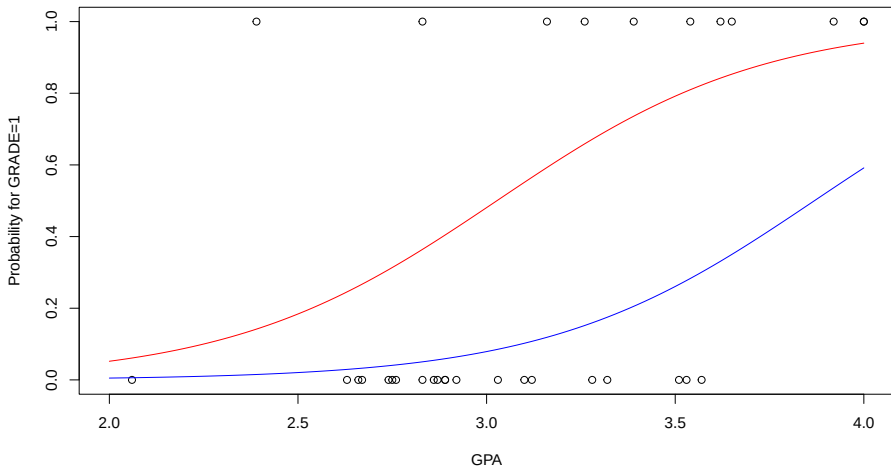
Table: Marginal effects

	Average marginal effect			Marginal effect at means		
	Logit	Probit	LPM	Logit	Probit	LPM
GPA	0.36	0.36	0.46	0.53	0.53	0.46
PSI	0.31	0.32	0.38	0.45	0.47	0.38
TUCE	0.01	0.01	0.01	0.02	0.02	0.01

Example: effectiveness of a new method of teaching

```
makePlot <- function() {
plot(d$GPA, d$GRADE,
     xlim=c(2, max(d$GPA)),
     xlab="GPA",
     ylab="Probability for GRADE=1")
curve( plogis( cbind( 1, x, mean(d$TUCE), 1 ) %*% coef(m1) ),
       add=TRUE, col = "red" )
curve( plogis( cbind( 1, x, mean(d$TUCE), 0 ) %*% coef(m1) ),
       add=TRUE, col = "blue" )
legend(0, .9, legend=c("PSI=1", "PSI=0"),
       col=c("red", "blue"), lty=1:1, cex=0.8)
}
```


Example: effectiveness of a new method of teaching



Example: effectiveness of a new method of teaching

3.5.2 Fitting a Bayesian model

Model statement

Data model:

$$\text{grade}_i \sim \text{Binomial}(1, p_i)$$

$$\text{logit}(p_i) = \beta_0 + \beta_1 \text{gpa}_i + \beta_2 \text{tuces}_i + \beta_3 \text{psi}_i$$

Parameter model:

$$\beta_k \sim \text{Normal}(0, 5), \quad k = 0, \dots, 3$$

Example: effectiveness of a new method of teaching

Estimate posterior distribution using MCMC

Estimation using R package `rstanarm`

```
library(rstanarm)
my_prior <- normal(0, 5, autoscale=FALSE)
m4 <- stan_glm(GRADE ~ GPA + TUCE + PSI,
               data = d,
               family = binomial(link = "logit"),
               prior = my_prior,
               prior_intercept = my_prior,
               seed = 463728, iter=4000)
```

Example: effectiveness of a new method of teaching

```
## stan_glm
## family:      binomial [logit]
## formula:     GRADE ~ GPA + TUCE + PSI
## observations: 32
## predictors:  4
## -----
##              Median MAD_SD
## (Intercept) -14.60    5.14
## GPA          3.06     1.30
## TUCE         0.12     0.15
## PSI          2.53     1.12
##
## -----
## * For help interpreting the printed output see ?print.stanreg
## * For info on the priors used see ?prior_summary.stanreg
```

Example: effectiveness of a new method of teaching

```
posterior_interval(m4, prob = 0.90, pars = c("GPA", "TUCE", "PSI"))

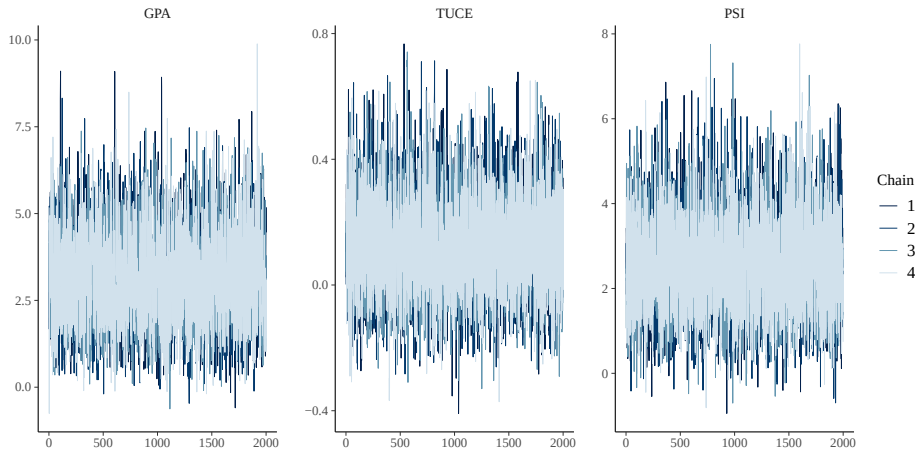
##              5%              95%
## GPA      1.1985026  5.4075364
## TUCE    -0.1127898  0.3933475
## PSI      0.8516180  4.5224418
```

After seeing the data, we believe that there is a 0.9 probability that the coefficient on PSI is between 0.85 and 4.52.

Example: effectiveness of a new method of teaching

Model checking

```
plot(m4, "trace", pars = c("GPA", "TUCE", "PSI"))
```



Example: effectiveness of a new method of teaching

Predictive comparisons

At means: we compare two individuals with $PSI=1$ and with $PSI=0$, assuming average values for GPA and TUCE.

```
plogis( cbind(1, mean(d$GPA), mean(d$TUCE), 1 ) %*% coef(m4) ) -
  plogis( cbind(1, mean(d$GPA), mean(d$TUCE), 0 ) %*% coef(m4) )

##           [,1]
## [1,] 0.4533935
```

Average comparison:

```
d$delta <- plogis( cbind(1, d$GPA, d$TUCE, 1 ) %*% coef(m4) ) -
  plogis( cbind(1, d$GPA, d$TUCE, 0 ) %*% coef(m4) )
round(mean(d$delta), 2)

## [1] 0.35
```

Example: effectiveness of a new method of teaching

Prediction

We inspect the posterior predictive distribution at different levels of the predictors.

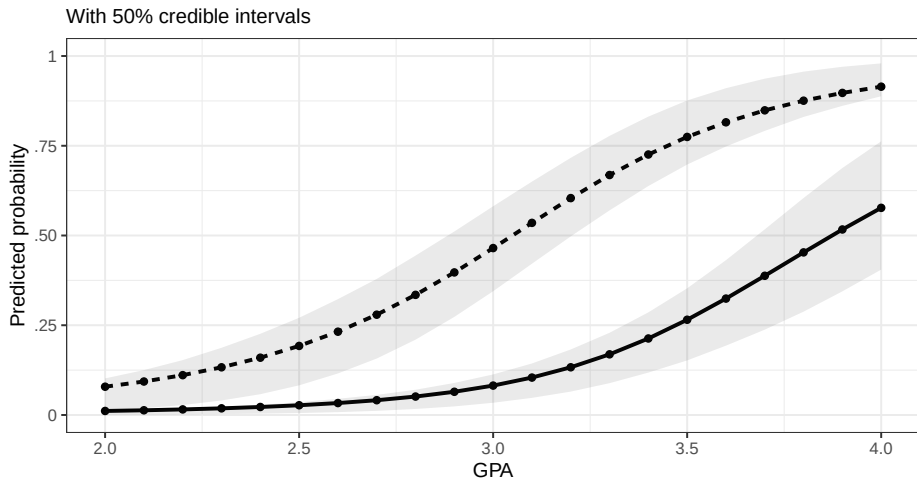
```
newdata0 <- data.frame(GPA = seq(2,4,0.1), TUCE=mean(d$TUCE), PSI=0)
newdata1 <- data.frame(GPA = seq(2,4,0.1), TUCE=mean(d$TUCE), PSI=1)
newdata <- rbind(newdata0, newdata1)
p_preds <- posterior_linpred(m4, newdata=newdata, transform=TRUE)

## Instead of posterior_linpred(..., transform=TRUE) please call
posterior_epred(), which provides equivalent functionality.

p_pred <- apply(p_preds, 2, mean)
p_pred_lb <- apply(p_preds, 2, quantile, probs=0.25)
p_pred_ub <- apply(p_preds, 2, quantile, probs=0.75)
newdata <- cbind(newdata, p_pred, p_pred_lb, p_pred_ub)
```


Example: effectiveness of a new method of teaching

```
## Warning: It is deprecated to specify 'guide = FALSE' to remove a  
guide. Please use 'guide = "none"' instead.
```



References

- Cameron, C. A. and Trivedi, P. K. (2005). *Microeconometrics. Methods and applications*. Cambridge University Press, Cambridge.
- Greene, W. H. (2011). *Econometric Analysis*. Prentice Hall International, Upper Saddle River, New Jersey.
- Winkelmann, R. and Boes, S. (2006). *Analysis of Microdata*. Springer, Berlin, Heidelberg.